Seonghyun Jin

Integrated M.S./Ph.D. Student | KAIST Graduate School of AI

jinotter3@kaist.ac.kr | jinotter@gmail.com | jinotter3.github.io | github.com/jinotter3 | Google Scholar

RESEARCH INTERESTS

Reconstructing and generating the real 3D world; streaming 3D reconstruction; geometric perception; Geometric Positional encoding; camera-controlled video generation.

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) Mar. 2025 – Present
Integrated M.S./Ph.D. in Artificial Intelligence Daejeon, Republic of Korea

- Advised by Prof. Jong Chul Ye at BISPL.

Seoul National University (SNU) Mar. 2020 – Feb. 2025
B.S. in Computer Science; B.A. in Economics, College of Liberal Studies Seoul, Republic of Korea

- GPA: 3.68/4.3, *Cum Laude*.

PUBLICATIONS

P: preprint, C: conference, W: workshop. *Equal contribution. Citations: Google Scholar, Sep. 11, 2026.

[P5] FastOPD: On-Policy Distillation for Lightweight VLA Deployment
Yoojin Oh*, Jeongsol Kim*, Yeonwoo Seo, Jangho Park, **Seonghyun Jin**, Sunwoo Park, Youngmin Kim, Youngjun Jun, Kyumin Choi, Jong Chul Ye
arXiv preprint, 2026 | **arXiv** | **PDF** | **Project**

[P4] DriftOPD: Sequence-Level Reverse-KL Distillation for One-Step VLA Policies
Youngjun Jun, Kyumin Choi, Youngmin Kim, **Seonghyun Jin**, Sunwoo Park, Jangho Park, Jong Chul Ye
arXiv preprint, 2026 | **arXiv** | **PDF** | **Project**

[P3] Adjoint Guidance Flow: Amortized Critic Guidance for VLA Policies
Jeongsol Kim, Youngjun Jun, Kyumin Choi, Youngmin Kim, **Seonghyun Jin**, Sunwoo Park, Jangho Park, Kwanyoung Kim, Jong Chul Ye
arXiv preprint, 2026 | **arXiv** | **PDF**

[P2] Don't Throw Away the Tail: Action Upcycling for Policy Acceleration
Taesung Kwon*, Jangho Park*, Sunwoo Park, Youngmin Kim, **Seonghyun Jin**, Youngjun Jun, Kyumin Choi, Jong Chul Ye
arXiv preprint, 2026 | **arXiv** | **PDF** | **Project**

[P1] Dynamic Manipulation with World-Action Models via Counterfactual Planning
Sunwoo Park*, Wonbin Lee*, **Seonghyun Jin***, Youngmin Kim*, Jangho Park, Jong Chul Ye
arXiv preprint, 2026 | **arXiv** | **PDF** | **Project**

[C3] CRePE: Curved Ray Expectation Positional Encoding for Unified-Camera-Controlled Video Generation
Seonghyun Jin*, Youngmin Kim*, Sunwoo Park*, Jong Chul Ye
NeurIPS 2026. Also accepted to 3D in the Era of World Models, ECCV 2026 Workshop | **arXiv** | **PDF** | **Code** | **Project**

[C2] FILT3R: Latent State Adaptive Kalman Filter for Streaming 3D Reconstruction
Seonghyun Jin, Jong Chul Ye
ECCV 2026 | **arXiv** | **PDF** | **Code** | **Project** | 2 citations

[C1] AI-Based Mental Health Assessment for Adolescents Using Their Daily Digital Activities
Do Hyung Kim, Joonsung Lee, Taehwi Lee, Soeun Baek, **Seonghyun Jin**, HaEun Yoo, Youngeun Cho, Seonghyeon Park, Kwangsu Cho, Chang-Gun Lee
IEEE DSAA, 2024 | **PDF** | 10 citations

ACADEMIC EXPERIENCE

RUBIS Lab, Seoul National University Sep. 2022 – Jun. 2024
Undergraduate Research Intern Seoul, Republic of Korea

- Contributed to adolescent mental health digital phenotyping research through feature engineering and data collection support; co-authored an IEEE DSAA 2024 publication.

LANGUAGES

Korean (native); English (fluent, TEPS 538, TOEIC 980).

TECHNICAL SKILLS

Programming Languages Python, C, C++
Deep Learning Frameworks PyTorch
Computer Vision Multi-view geometry, 3D reconstruction, camera pose and depth estimation
Training Experience Deep learning training, ablation studies, experiment management, evaluation pipelines

REFERENCE

Professor Jong Chul Ye
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